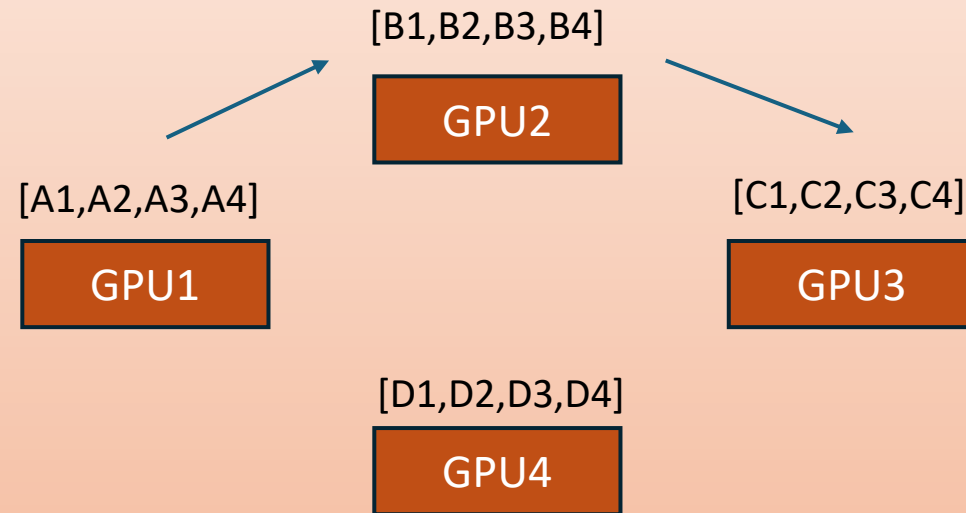
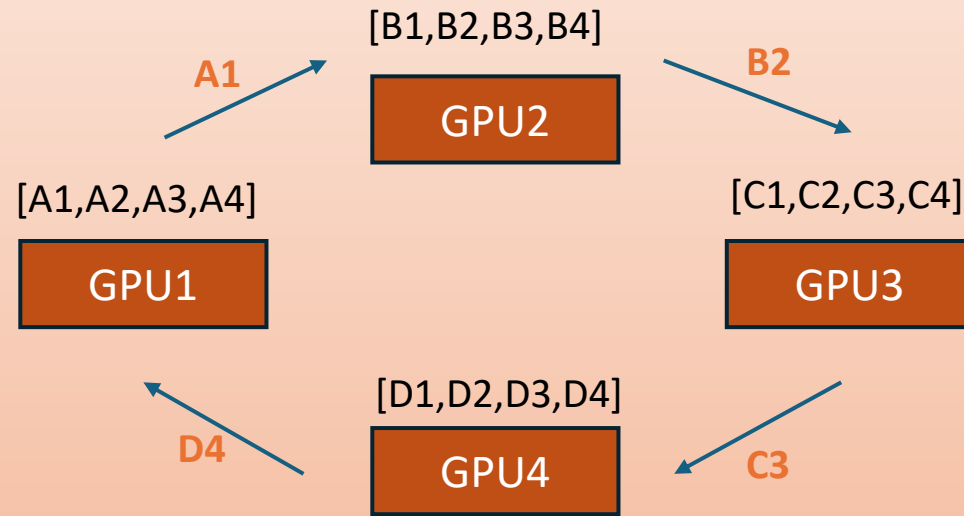


Ring All-Reduce example



1. Initial State: Each GPU splits its gradient into 4 chunks (1 per GPU).

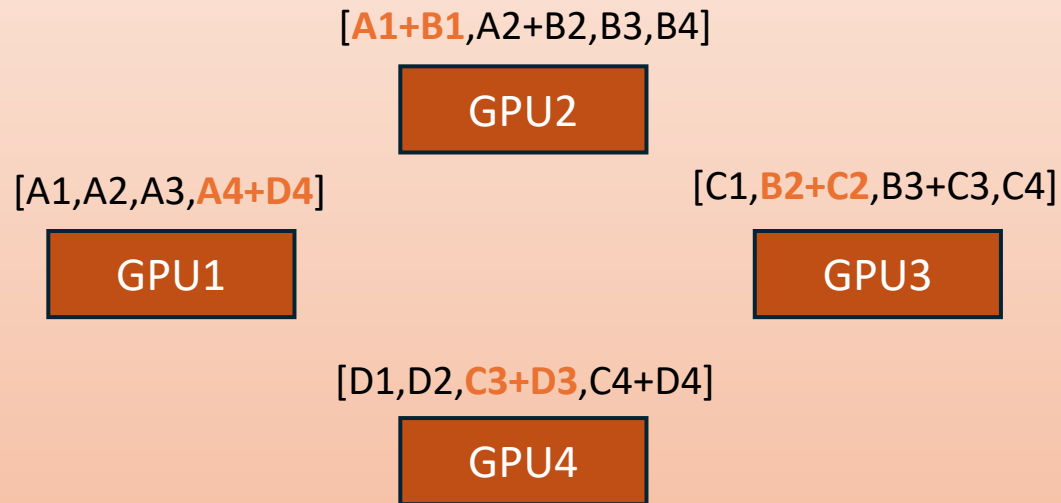
Ring All-Reduce example



"Share" Step

2. Ring Communication (Reduce – Scatter) phase : All GPUs send chunks to their neighbors. The chunks are added (reduced).

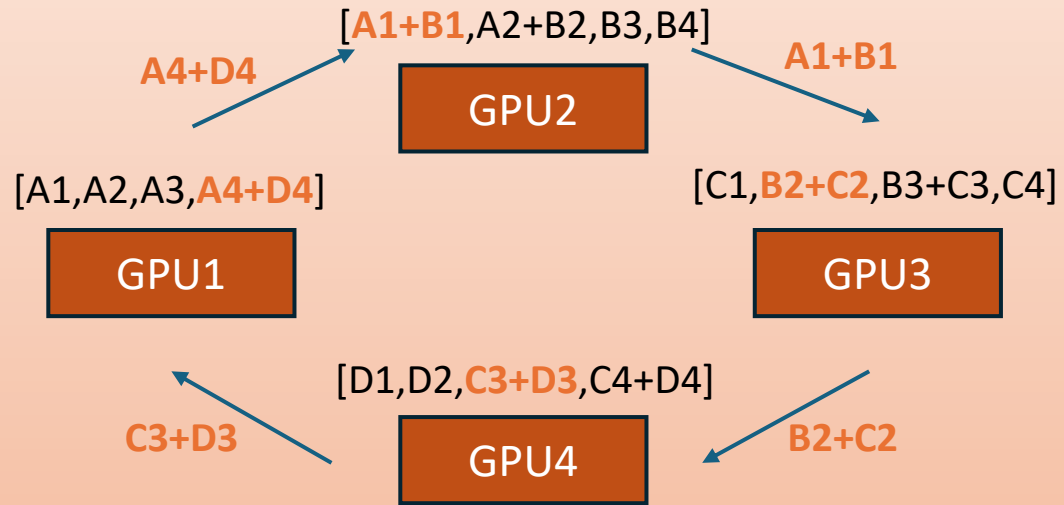
Ring All-Reduce example



"Reduce" Step

2. Ring Communication (Reduce – Scatter)
phase : All GPUs send chunks to their neighbors. The chunks are added (reduced).

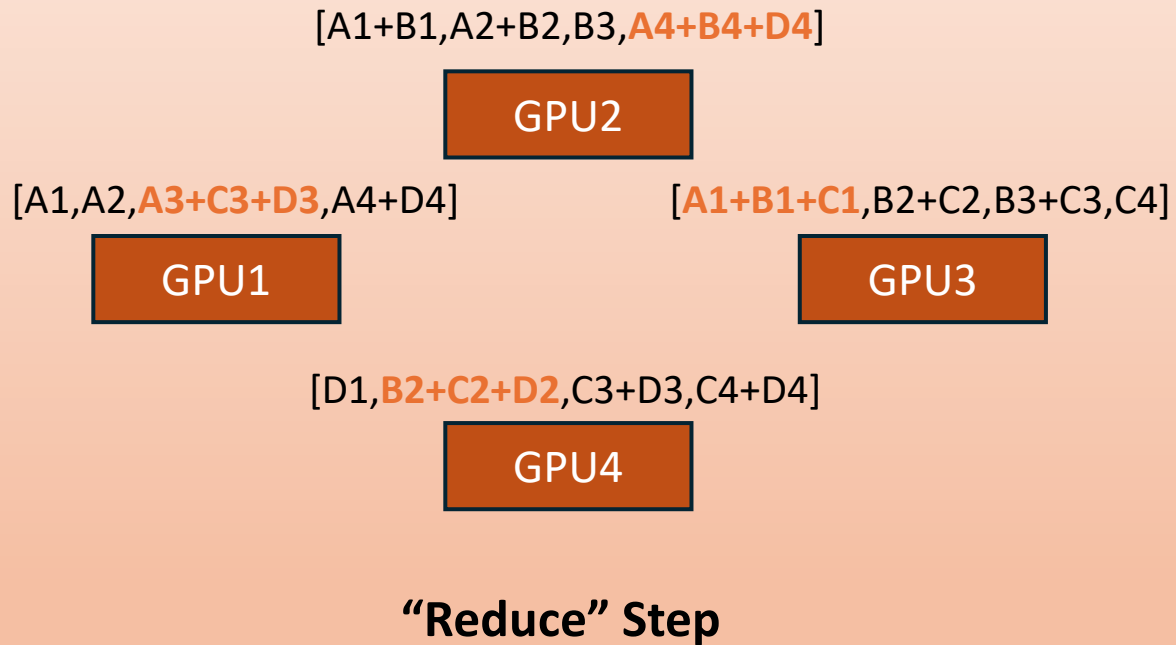
Ring All-Reduce example



"Share" Step

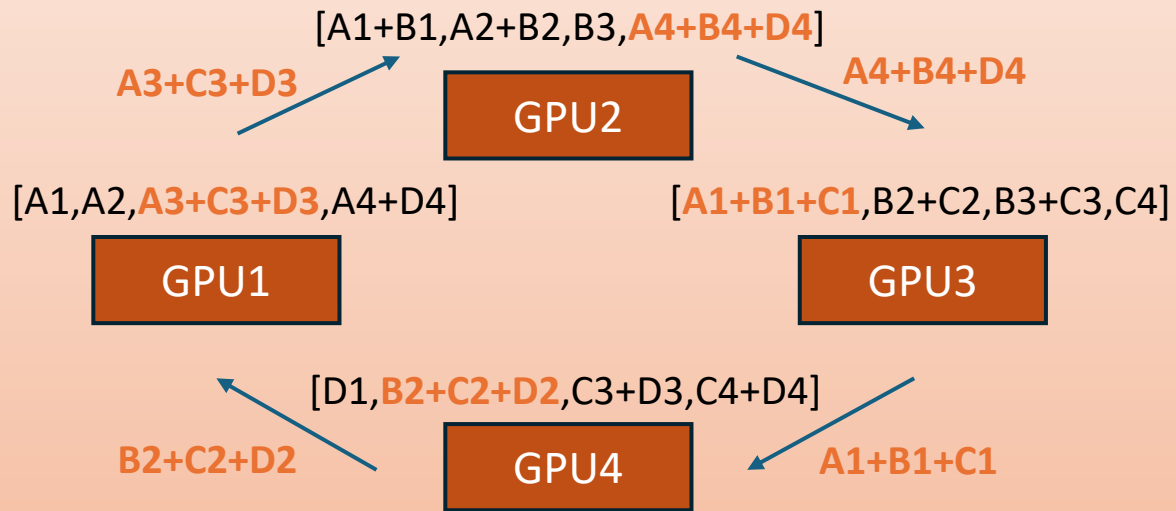
2. Ring Communication (Reduce – Scatter) phase : All GPUs send chunks to their neighbors. The chunks are added (reduced).

Ring All-Reduce example



2. Ring Communication (Reduce – Scatter)
phase : All GPUs send chunks to their neighbors. The chunks are added (reduced).

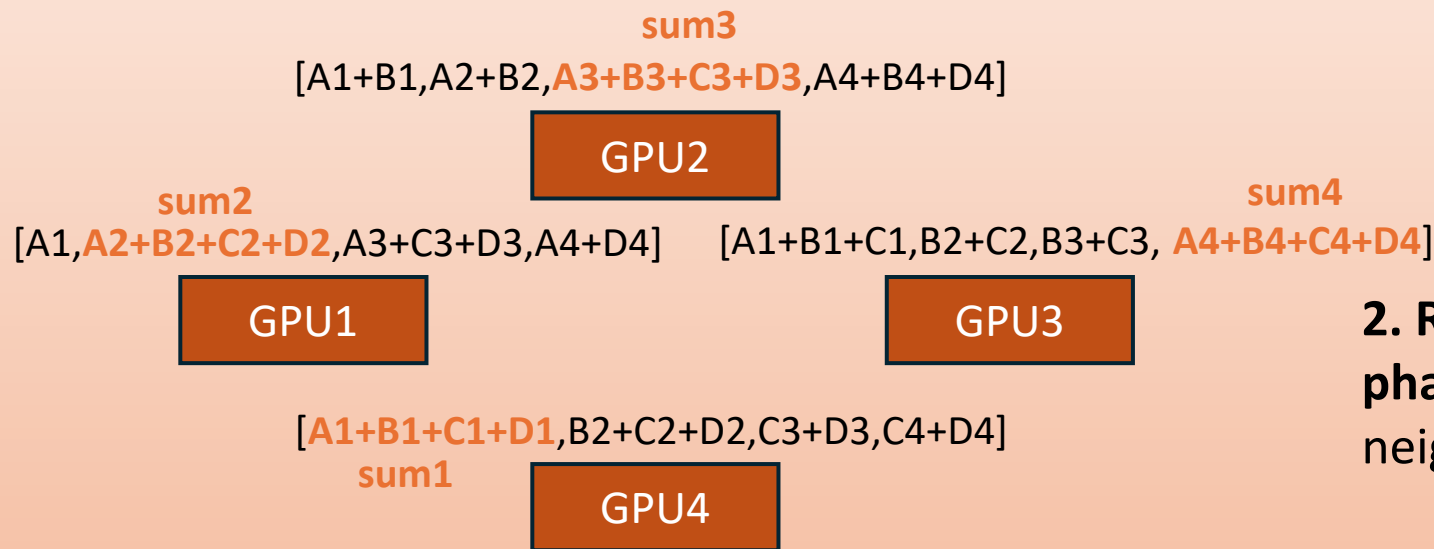
Ring All-Reduce example



“Share” Step

2. Ring Communication (Reduce – Scatter) phase : All GPUs send chunks to their neighbors. The chunks are added (reduced).

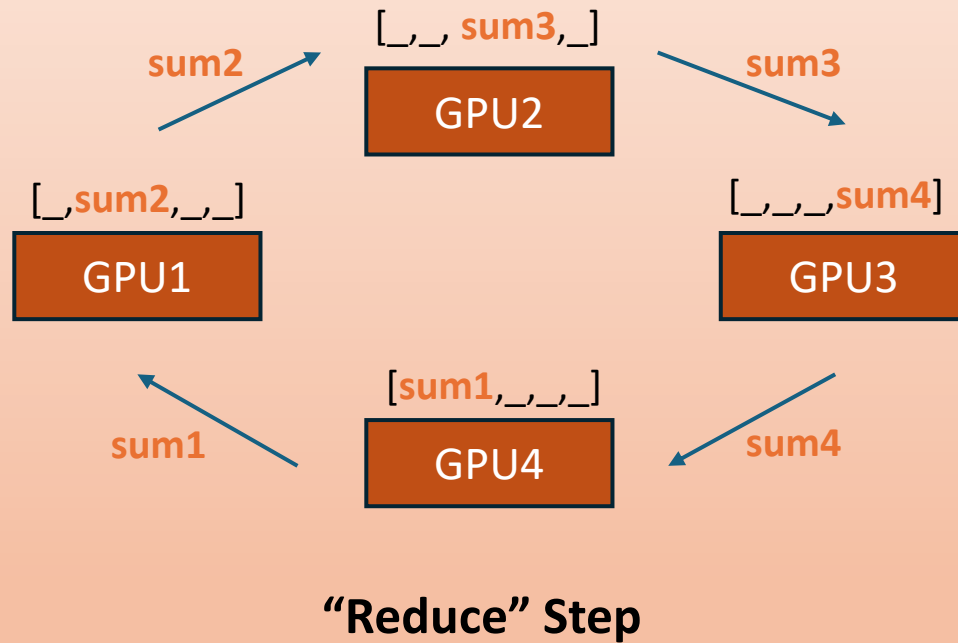
Ring All-Reduce example



“Reduce” Step

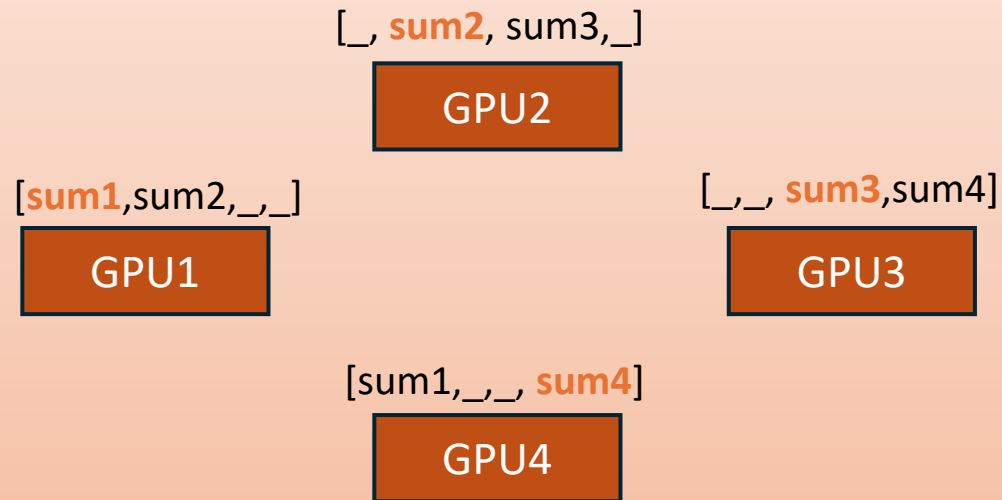
2. Ring Communication (Reduce – Scatter)
phase: All GPUs send chunks to their neighbors. The chunks are added (reduced).

Ring All-Reduce example



3. All-gather (broadcast) phase: each GPU sends its fully summed chunk to its neighbor

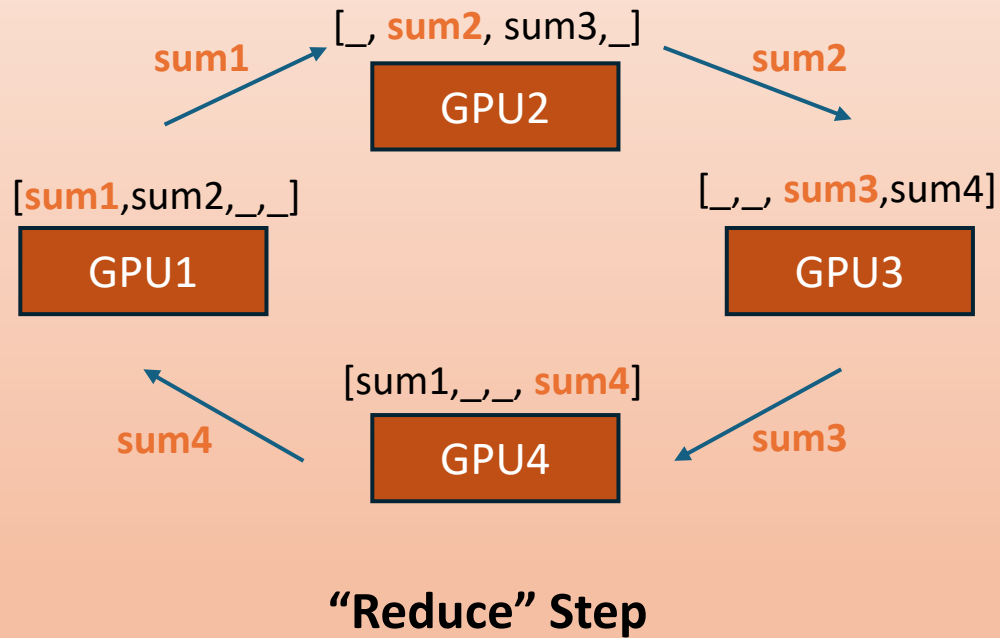
Ring All-Reduce example



“Reduce” Step

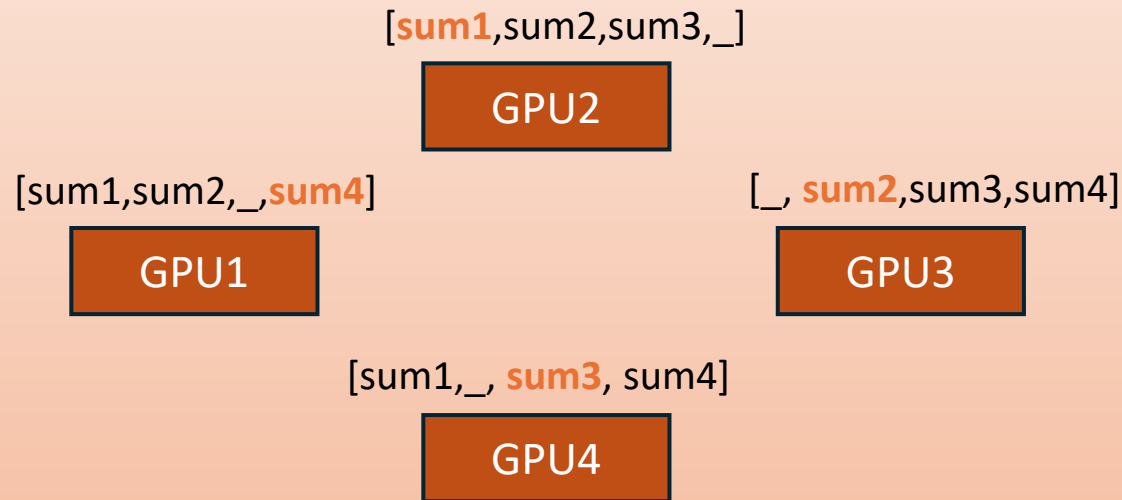
3. All-gather (broadcast) phase: each GPU sends its fully summed chunk to its neighbor

Ring All-Reduce example



3. All-gather (broadcast) phase: each GPU sends its fully summed chunk to its neighbor

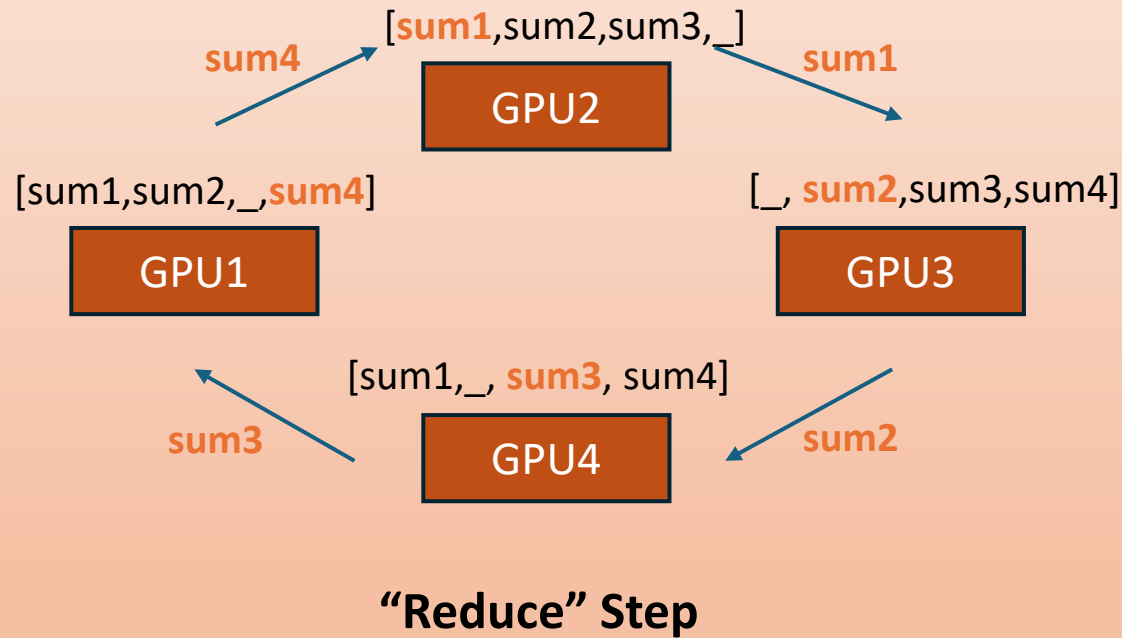
Ring All-Reduce example



“Reduce” Step

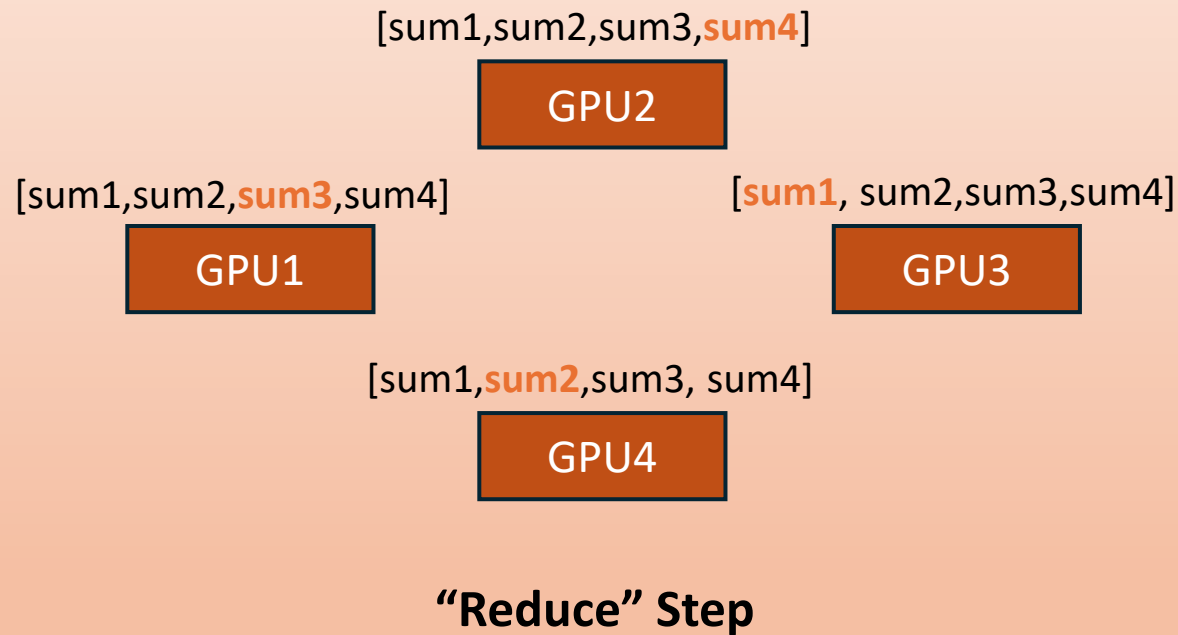
3. All-gather (broadcast) phase: each GPU sends its fully summed chunk to its neighbor

Ring All-Reduce example



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Ring All-Reduce example



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Topology-Aware & Hierarchical All-Reduce; Libraries

- Libraries:
 - NCCL (GPU-centric), Gloo (CPU/GPU), MPI (HPC)
- Algorithms:
 - Ring: bandwidth-optimal for large messages
 - Tree: lower latency for smaller messages
 - Reduce-scatter + all-gather: core of sharded training
- Hierarchical all-reduce:
 - Intra-node reduce-scatter (NVLink)
 - Inter-node reduce-scatter (IB/Ethernet)
 - Inter-node all-gather
 - Intra-node all-gather
- Practice: bucket sizes, algorithm selection per tensor size, topology hints